Queuing Theory: Part 5

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Queuing Decision Models

- Focus on **making optimal decisions** about managing queues and service systems to improve performance, reduce waiting times, and increase customer satisfaction.
- These models combine **queueing theory** with **decision-making tools** to guide how to design or control service systems.
- These models assist in
  - **Service Capacity Decisions:**

    How many servers should be deployed?

    When to add or remove service counters?

Queuing Decision Models

- **Queue Discipline Decisions:**
  - Which priority rules to use (e.g., First-Come-First-Served, priority queues)?
  - Should some customers be prioritized or given faster service?
- **Routing and Scheduling:**
  - How to assign arriving customers to different service channels or queues?
  - Scheduling server availability during peak and off-peak times.

Examples

- Hospital Emergency Room: Prioritizing patients by severity to minimize critical waiting.
- Manufacturing System: Deciding buffer sizes and processing stations to optimize throughput and minimize delays.

Queuing Decision Models

- **Cost Models**

Quantifies the total cost involved in operating a system. Assist in finding the most cost-effective way to design or operate a system. Cost components are

Operating costs: labor, equipment, utilities

Waiting costs: customer waiting time, penalties for delay

Service costs: cost of servers or resources used

Penalty costs: costs from poor service, lost customers, or system failures

- Example:

  In a queueing system, the cost model might include the cost of staffing more servers versus the cost incurred from customers waiting too long (leading to dissatisfaction or lost business). By modeling these costs, the optimal number of servers can be decided.

- **Example:** A single-server service center receives customers at a rate of 10 per hour (Poisson arrival). The average service rate is 12 per hour (exponential service). The customer waiting cost is Rs. 20 per customer per hour (in system) and server cost is Rs. 100 per hour. Determine

  (a) utilization factor

  (b) Expected number of customers in the system

  (c) Average time a customer spends in the system

  (d) Total expected cost per hour

- Given,

Arrival rate, λ = 10 customers/hour

Service rate, μ = 12 customers/hour

Waiting (in system) cost per customer/hour = Rs. 20

Server cost per hour = Rs. 100

(a) Utilization Factor

$$\rho = \frac{\lambda}{\mu} = \frac{10}{12} = 0.8333$$

(b) Expected Number of Customers in the System

$$L_s = \frac{\rho}{1-\rho} = \frac{0.8333}{1-0.8333} = 4.99$$

(c) Average Time in System

$$W = \frac{L}{\lambda} = \frac{4.99}{10} = 0.4999\,\text{hrs} = 29.94\ \text{mins}$$

(c) Waiting cost

Waiting Cost per hour = L × Cost per customer/hour

= 4.99 × 20 = Rs. 99.8

Total cost = Rs 99.8 + server cost

= Rs. 99.8 + Rs. 100

= Rs. 199.8/-

Queuing Decision Models (continued…)

- **Aspiration Level Model**

It is a type of decision-making model where the decision-maker sets a **target or desired level (aspiration level)** for certain performance measures such as waiting time, queue length or system cost.

Then it attempts to adjust system parameters (like service rate, number of servers, etc.) to meet or exceed that level.

Queuing Decision Models (continued…)

- **Aspiration Level:** A predefined acceptable limit for performance. For example, "average waiting time should not exceed 5 minutes".
- **Decision Variable:** Parameters under control (number of servers, service rate) that can be changed to reach the aspiration level.
- **Optimization is Goal-Oriented:** Unlike traditional cost models that aim to minimize total cost, aspiration level models seek to satisfy service level expectations. This may not be purely cost-optimal.

Uses of Aspiration Level Models:

In **service design** (banks, hospitals, call centers)

In **public service systems** where user satisfaction is prioritized

- **Example:** A manager wants to ensure that customers do not have to wait longer than 6 minutes on average. The arrival rate is 10 customers/hour. What should be the minimum service rate (μ) ?
- Solution: Given

λ = 10 customers/hour

W = 6 minutes = 0.1 hours (aspiration level for average time in system)

From queuing theory (M/M/1):

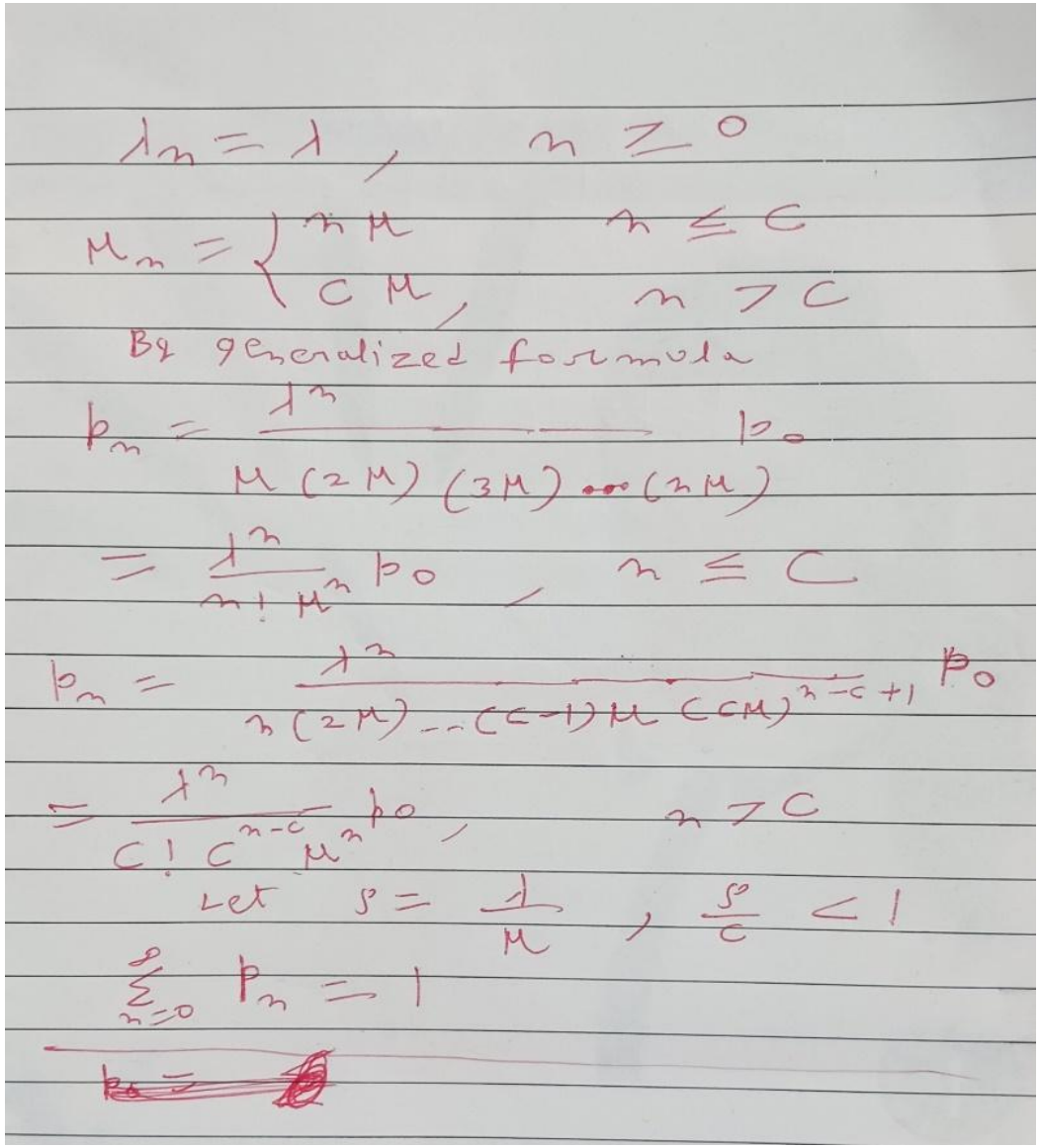
$$W = \frac{1}{\mu - \lambda}$$

$$\text{or } 0.1 = \frac{1}{\mu - 10}$$

$$\text{or } \mu = 20$$

To meet the aspiration level of 6 minutes maximum waiting, the service rate **μ must be at least 20 customers/hour**.

Multi-server Model



$$\lambda_0 p_0 = \mu_1 p_1$$

$$p_n = \left(\frac{\lambda_{n-1}\lambda_{n-2}\ldots\lambda_0}{\mu_n \mu_{n-1\ldots}\mu_1} \right) p_0, \; n = 1, 2, 3, \ldots..$$

$$p_0 = \left\{ \sum_{n=0}^{c-1} \frac{\rho^n}{n!} + \frac{\rho^c}{c!} \left(\frac{1}{1 - \frac{\rho}{c}} \right) \right\}^{-1}, \frac{\rho}{c} < 1.$$

$$L_q = \sum_{n=c+1}^{\infty} (n-c)p_n = \frac{\rho^{c+1}}{(c-1)!(c-\rho)^2} p_0$$

$$L_s = L_q + \rho$$

Apply Little's formula for finding out waiting time and time in system.

If the inter-arrival time is exponential and 10 customers per hour arrive in a bank, then the probability of no arrival of customer during a period of 10 minutes is_____.

Here, λ=10 per hour=1/6 per minute

$$P\{t \leq 10\} = 1 - e^{-10/6};$$

$$P\{t \geq 10\} = e^{-10/6} = 0.1889$$

By Poisson distribution:

$$p_n(t) = \frac{(\lambda t)^n e^{-\lambda t}}{n!}, \quad n = 0, 1, 2, 3.....$$

Hence,

$$p_0(10) = e^{-\lambda t} = e^{-10/6} = 0.1889$$

A bank has one teller. Customers arrive at a rate of 10 per hour (Poisson distribution), and the teller serves them at a rate of 12 per hour (exponential distribution). What is the average number of customers waiting in the queue?

- $\lambda = 10$ customers/hour
- $\mu = 12$ customers/hour
- $\rho = \lambda/\mu = 10/12 = 5/6$
- $L_q = \rho^2 / (1 - \rho) = (5/6)^2 / (1 - 5/6) = 25/6$ customer

- $L_s = L_q + \rho = 25/6 + 5/6 = 5$ customers
- $W_s = 5/\lambda = 0.5$ hour

Finite Waiting Line Model

- The Finite Waiting Line Model, also known as a finite queue model, is a type of queuing system where the number of potential customers or the system capacity is limited. Unlike infinite queue models where an unlimited number of customers can join the line, the finite model considers practical constraints such as limited space, limited customer population, or restricted system capacity.
- In some books you will get finite queuing tables to solve. You need three inputs: size of the population N, the service factor X and the number of parallel server M.
- X=(mean arrival rate per population level)/(mean arrival rate per population level+ service rate)
- Mean waiting time+(1/service rate)=Mean time in system
- Mean number of units being served= Mean system length–Mean queue length=H
- M–H=Mean number of servers idle